AI-based campus assistant application development

"i want to create an platform independent app for campus copilot (AI based) that utilises college data from various college websites and apps along with spam mail detection" *Generated by LeapSpace*

Campus Copilot AI App at a Glance

Key Findings Table

Topic	Key finding	Practical implication	Citations
Multi-source campus data	Campus assistants can ingest PDFs, notices, circulars, syllabi, images/posters, timetables, and API-fed data; OCR and retrieval are central to making these searchable and usable.	Start with a mixed ingestion pipeline rather than a chatbot-only design.	[1] [2]
End-to-end integration maturity	Evidence supports pieces of the pipeline—OCR, metadata transformation, deduplication, redaction, repository migration—but not one validated single architecture covering all campus source types together.	Build modularly; do not assume one “standard” campus copilot stack already exists.	[3] [4] [5] [6]
App/platform strategy	Shared frontend codebases plus backend APIs are the strongest pattern for platform-independent delivery, but Flutter, React Native, and PWA each involve tradeoffs in performance, learning curve, web compatibility, and maintenance.	Use shared UI where possible, but keep intelligence and integrations in backend services.	[7] [8] [9] [10]
Campus copilot architecture	RAG is the dominant approach for campus Q&A because it grounds responses in institutional sources and reduces hallucination relative to unsupported generation.	Make retrieval and citations first-class features.	[1] [11] [12] [13]

Topic	Key finding	Practical implication	Citations
Retrieval design	Stronger systems increasingly use multi-stage routing: FAQ matching, lexical or dense retrieval, reranking, and sometimes graph or hierarchical reasoning.	Route simple queries cheaply before invoking heavier LLM steps.	[14] [15] [16] [17] [18]
Assistant capabilities	Best-supported functions are campus FAQ/Q&A, document-grounded answers, and summarization; evidence for notifications and timetable reminders is much weaker.	MVP should focus on search, Q&A, and summarization before advanced reminder automation.	[1] [11] [15] [19] [20] [21] [22] [23]
Spam detection	Spam filtering has a more mature evidence base than campus harmonization, with strong results from Random Forest, XGBoost, Naive Bayes, logistic regression, ensembles, and newer transformer-style models.	Use classical ML baselines first; add deeper models only after benchmarking.	[24] [25] [26] [27] [28] [29] [30] [31]
Privacy and compliance	Least-privilege data collection, API-based access, PII redaction, and header-only spam detection are recurring privacy-preserving patterns.	Build privacy-by-design into ingestion and email processing.	[5] [32] [33] [34] [35] [36]

Direct Answer

A practical way to build your platform-independent AI campus copilot is as **two interoperable subsystems**:

> a **campus knowledge subsystem** that ingests official college data from websites, PDFs, notices, posters, timetables, and authorized app-backed APIs, normalizes it, and serves it through a **RAG-based assistant with citations**; and
> a **spam-mail detection subsystem** that classifies suspicious academic emails using proven ML baselines such as **Random Forest or XGBoost**, with **header-only features** as a privacy-preserving fallback when message bodies are sensitive or unavailable [1] [11] [26] [29] [35] [36] [37].

For an MVP, prioritize:

> official website/PDF/API ingestion,
>
> search + grounded Q&A,

document summarization,

a separate spam-classification service,

backend APIs shared across web, Android, and iOS clients [1] [12] [19] [37].

Avoid starting with complex reverse engineering of college apps, heavy multi-tenant identity, or fully automated reminder workflows, because evidence for those deployment details is relatively weak compared with evidence for retrieval-based assistants and spam classification [11] [15] [21] [22] [23] [33] [38] [39] [40].

Confidence level: High - The core recommendation is strongly supported across the synthesis and multiple topic clusters, though not by one single end-to-end reference system [1] [3] [4] [5] [11] [26] [29].

Study Scope

This report covers research summarized in the supplied evidence on:

- campus data ingestion from websites, documents, OCR sources, and app-backed services [1] [32] [41];
- cross-platform frontend and backend architecture patterns for web, Android, iOS, and shared services [7] [8] [12] [37];
- AI assistant design for campus search, Q&A, summarization, and multi-stage retrieval [1] [11] [14] [15];
- spam-mail detection models, features, and evaluation methods from 2019-2025-style evidence in the provided set [24] [25] [26] [29];
- privacy, oversight, and access-control considerations relevant to educational deployments [42] [43] [44] [45].

Assumptions & Limitations

The supplied evidence does **not** contain one validated production architecture combining heterogeneous campus ingestion, website/LMS/mobile deployment, authentication, rate limiting, analytics, and spam detection in one system [3] [4] [5] [6] [38] [39] [40].

There is limited direct evidence for notifications and timetable reminders compared with Q&A and summarization [11] [15] [21] [22] [23].

University-specific spam benchmarks comparing all major model families side by side were not found in the supplied material [26] [30] [31] [46] [47].

Suggested Further Research

Compare **Flutter vs React Native vs PWA** for your exact delivery targets, especially web compatibility and push behavior [7] [8] [9] [10].

Prototype a **tiered ingestion strategy**: official APIs first, structured web next, then PDFs/OCR, then authorized app endpoints [1] [32] [33].

Benchmark **FAQ matching vs dense RAG vs reranked RAG** on real campus questions using retrieval and answer-quality metrics [14] [15] [18] [48].

Build a university-specific spam dataset and compare **RF, XGBoost, NB, logistic regression, and transformer models** under privacy constraints [26] [30] [31].

Study institutional policies for **role-based access, PII redaction, and human review workflows** before integrating student data [5] [44] [45].

Full Report

Introduction and Problem Framing

A campus copilot is best understood as a unified assistant that helps students and staff query institutional information, navigate resources, summarize documents, and potentially support planning tasks such as schedules or service access [1] [11] [49]. The strongest evidence supports assistants that answer administrative and academic questions by grounding responses in campus documents and knowledge bases rather than relying on unguided generation [11] [50] [51].

For your goal, “platform independent” is not mainly an AI problem; it is an **architecture problem**. The evidence favors a design where the AI logic, ingestion, retrieval, and security live in shared backend services, while the user interface is delivered through one or more shared frontend approaches across web, Android, and iOS [7] [12] [37]. At the same time, spam-mail detection should be treated as a related but distinct subsystem because its data, models, metrics, and privacy requirements differ from campus-document Q&A [26] [29] [37].

Confidence level: High - The separation between assistant and spam-detection concerns is consistently supported across the synthesis and source clusters [1] [11] [26] [29].

Foundational Context

What the system must actually do

Your target product combines four technical functions:

Acquire campus data from heterogeneous sources such as websites, notices, PDFs, images, timetables, and app-backed endpoints [1] [32].

Normalize and index that data so a retrieval system can search it reliably across departments and formats [4] [5].

Serve grounded AI answers through retrieval-augmented generation, ideally with citations and routing to live tools or APIs where needed [1] [11] [14].

Filter spam mail using a separate classification pipeline with privacy-sensitive feature handling [26] [34] [35].

Why RAG matters here

RAG combines retrieval from a document or knowledge index with answer generation, which improves factuality and specificity over purely parametric generation on knowledge-intensive tasks [13]. In campus settings, this matters because policies, notices, schedules, and department information change frequently; retrieval helps answers stay tied to current institutional sources [1] [11] [50].

Why a modular system matters

Taken together, the findings suggest that campus knowledge assistance and spam filtering should share infrastructure such as APIs, logging, and privacy controls, but remain operationally separate because they use different pipelines and success criteria [11] [26] [29] [37]. This is an inference from the evidence rather than a directly validated single-system architecture.

Confidence level: High - The underlying parts are well evidenced; the combined modular recommendation is a synthesis across sources [1] [11] [26] [37].

Campus Data Acquisition and Normalization

Source types and ingestion channels

Campus-oriented systems can ingest a wide range of source formats: PDFs, circulars, notices, syllabi, images or posters, timetables, and API-fed event or schedule data [1]. OCR is especially important for turning scanned or image-based material into searchable text [1] [3]. Structured support systems also show that location and navigation information can be served from curated knowledge bases through web interfaces [52] [53] [54].

For app-backed or portal-backed services, the evidence supports API-based acquisition rather than direct database access. One study reports using application APIs to obtain, interpret, format, and store campus records without touching the source database directly [32]. Traffic capture of web-application APIs has also been used to identify request paths, parameters, and response payloads for later normalization [41].

A practical ingestion hierarchy follows directly from this evidence:

Tier 1: official APIs and feeds [32];

Tier 2: structured public web pages and portal pages within authorized scope [33];

Tier 3: PDFs, scanned notices, posters, and image-heavy content through OCR [1] [3];

Tier 4: authorized app-derived endpoints only when institutionally permitted [32] [33].

This tiered approach is an inference, but it aligns closely with the privacy and maintainability patterns in the supplied evidence [5] [33].

Normalization, metadata, and deduplication

The evidence for full campus harmonization is fragmented, but several pipeline components are well supported. A repository migration case study used metadata scraping, transformation, cleanup, and schema updates at institutional scale [4]. A grievance pipeline demonstrated PII redaction, duplicate detection using MinHash-LSH and cross-encoder entailment, calibration, and HTTP API exposure [5]. Another pipeline combined OCR with semantic clustering and summarization for scanned and textual archives, which approximates normalization and organization over messy institutional documents [3].

What is missing is a comparative literature on schema design, entity resolution, taxonomy creation, and update policies specific to university portals and department websites [55] [56] [57] [58] [59] [60]. So, your schema will likely need to be engineered from first principles.

A minimal normalized campus schema should therefore capture:

- source URL or source system,
- document type,
- department or office,
- date or validity period,
- title and extracted body text,
- access level,
- freshness timestamp,
- duplicate group or canonical record link [4] [5].

That schema proposal is an inference based on the documented metadata transformation and deduplication needs [4] [5].

Practical constraints

The evidence strongly favors compliant acquisition. Studies describe scraping or API use that stays within allowed restrictions and manual-access scope, explicitly avoiding unauthorized access or security bypass [33]. When direct access is unavailable, researchers use web scraping, platform APIs, or dedicated tooling rather than database intrusion [61]. This implies that reverse engineering campus apps may be technically possible in some contexts but should not be your default product strategy [32] [33].

Confidence level: Medium-High - Ingestion methods are well supported, but normalization policy for campus-specific mixed sources remains underdeveloped in the evidence [55] [60].

Platform-Independent System Architecture

Frontend options

The evidence does not identify one universal winner. Flutter showed strong Android performance among single-codebase frameworks, while React Native had the lowest learning curve and Xamarin the easiest setup in one comparison [7]. PWAs are described as promising for unified cross-platform development, but not yet a complete “technology unifier” [8]. Flutter-focused studies also note ongoing security and compatibility problems, especially on web targets, which can increase maintenance effort [9] [10].

So the platform choice should follow delivery priorities:

Flutter if mobile performance is central, especially Android-heavy deployment [7];

React Native if team familiarity and faster onboarding matter more [7];

PWA/web if instant access and browser availability are top priorities, while accepting capability limits [8].

Backend and service layer

The backend should be the stable core. Shared-service patterns are supported by evidence showing university assistants deployed across web channels and LMS-like environments through centralized chatbot infrastructure [12]. For assistant architecture choice, privacy, scalability, and compliance are explicit selection criteria in educational chatbot design [37].

A strong backend decomposition is:

authentication and role handling,

ingestion jobs and source connectors,

search and vector retrieval,

answer orchestration and citation assembly,

live tool/API connectors for timetable or event data,

spam-classification API,

audit logging and analytics [1] [12] [37].

The exact production details for authentication, rate limiting, analytics, and multi-tenant support are notably absent in the retrieved evidence, so those parts must be specified as engineering design rather than literature-backed standards [38] [39] [40].

Offline, latency, and AI-assisted development

For mobile AI patterns, on-device designs offer stronger offline performance and privacy but cost more in storage, while client-server hybrids provide more flexibility but introduce network-dependent latency [62]. For your use case, most retrieval and campus indexing should probably remain server-side because source freshness matters; only lightweight caching or local UI behavior clearly benefits from offline mode [1] [62]. This is an inference from the deployment tradeoff evidence.

AI coding tools can accelerate development, but they are weaker on multi-layer APIs and architecture-heavy work than on client-side coding, with common errors caused by lost context across files and layers [63]. Human review remains essential [64].

Confidence level: High - Shared frontend plus backend APIs is well supported, though fine-grained operational architecture is underreported [7] [12] [37] [38].

Assistant Capabilities and Retrieval Design

Best-supported features

The strongest evidence supports:

- FAQ automation and semantic search [11] [12];
- grounded campus Q&A over documents and institutional sources [1] [50] [51];
- curriculum- or knowledge-base-specific help [65] [66];
- document summarization, notes, and reading support [19] [20].

Some systems also explicitly address schedules and department resources, but the evidence on reminder workflows or notification pipelines is limited [11] [15] [21] [22] [23] [49].

Recommended orchestration design

The evidence consistently points toward **multi-stage retrieval** rather than a single chat step. A high-value campus assistant can route queries through:

- FAQ or intent matcher for common policy questions [14] [67];
- lexical or dense retrieval for document lookup [15];
- cross-encoder reranking for answer grounding [15];
- live API/tool calls for timetable or event data [1];
- summarization for long notices or PDFs [19].

This staged design is supported by educational RAG studies reporting metrics such as Precision@K, Recall, F1, BLEU, and ROUGE-L, plus efficient real-time processing with reranking and embeddings [18] [48] [67]. Hallucination reduction is also reported when retrieval grounds generation in evidence [67] [68].

Citation-aware answers

Source attribution is especially important in an institutional context because users need to verify policies and announcements [1] [2]. The evidence supports systems that cite source documents in responses, especially document-QA or RAG-style tools [1] [19].

Confidence level: High - RAG and staged retrieval are among the most consistently supported findings in the entire evidence set [1] [11] [14] [15].

Spam-Mail Detection for Academic Email Streams

Feature engineering

Spam detection studies in the supplied material repeatedly use tokenization, stemming, stopword removal, TF-IDF, word entropy, normalized length, and metadata or header features [26] [34] [69] [70]. When email bodies are unavailable or sensitive, header-only pipelines remain viable. One framework uses just five header fields, while another extracts 48 header-based features and reports strong accuracy and ROC area [35] [36].

This makes header-first detection particularly suitable for educational institutions with privacy constraints [34] [35] [36].

Model choices

The evidence base is much stronger here than for campus harmonization. Reported models include Random Forest, XGBoost, Naive Bayes, logistic regression, KNN, SVM, ensemble voting, transformer-style systems, and other deep or hybrid approaches [26] [27] [28] [69] [71].

For practical baselines:

- **Random Forest** repeatedly performs strongly in academic or email-like settings [24] [25] [30] [31];
- **XGBoost** also reports strong results, including 97% accuracy and 0.97 AUC in one study [26];
- **Ensembles** can outperform single models in some datasets [29] [31];
- **Transformers** are promising, but the supplied evidence does not establish them as universally superior on university email specifically [26] [29] [30] [31].

Evaluation and deployment

Reported metrics commonly include accuracy, precision, recall, F1, ROC-AUC, and sometimes execution-time or false-positive/false-negative behavior [26] [29] [69]. Methodological baselines in the evidence also emphasize ensemble learning, class imbalance handling, and cost-sensitive evaluation, which are important because false positives in academic email can be costly [72] [73] [74].

A sound deployment pattern is:

score incoming mail,

quarantine high-risk messages,

explain likely spam indicators,

allow user/admin feedback for retraining [45] [73].

The “explain + quarantine + feedback” workflow is an inference from the methodological and secure-processing evidence rather than a directly reported single workflow [45] [73].

Confidence level: High - The model and feature guidance is strongly grounded across multiple studies [24] [25] [26] [29] [35] [36].

Privacy, Access Control, and Governance

Educational AI literature stresses responsible use, fact-checking, policy controls, and human oversight because of inaccuracy, bias, privacy, plagiarism, and legal concerns [42] [43] [44] [75]. In software implementation work, human review is also necessary to preserve code quality and architectural integrity when AI tools are used [63] [64].

For your system, privacy-by-design means:

use official APIs or authorized scraping where possible [32] [33];

redact or minimize PII in ingestion pipelines [5];

separate role-restricted data from public institutional content [37] [44] [76];

prefer header-only or secure API-based email processing when bodies are sensitive [34] [35] [45].

Confidence level: Medium-High - Governance priorities are clear, but detailed student-data integration policies are not deeply specified in the supplied technical evidence [44] [45].

Recommended Product Roadmap

MVP

Build first:

shared backend APIs [12] [37];

ingestion for official websites, PDFs, and approved APIs [1] [32];

normalized search index with metadata [4];

RAG-based Q&A with citations [1] [11];

document summarization [19];

separate spam-classification service using Random Forest and XGBoost baselines [24] [26].

Phase 2

Add:

reranking and query routing [14] [15];

richer deduplication and freshness logic [5];

role-based restricted content access [37] [76];

header-only spam mode for privacy-sensitive deployments [35] [36].

Phase 3

Consider:

deeper app integrations where explicitly authorized [32] [33];

more advanced live tool calls for timetable/event data [1];

user feedback loops and administrative analytics [12] [45].

Confidence level: High - The staged roadmap aligns with where the evidence is strongest and avoids areas where implementation support is weak [1] [11] [26] [38].

Conclusion

The supplied evidence supports building your app as a **modular, privacy-first, platform-independent system**: a shared frontend experience backed by centralized APIs; a campus knowledge layer using compliant multi-source ingestion plus RAG; and a separate spam-mail detector using proven ML baselines and privacy-preserving feature sets [1] [11] [12] [26] [35] [37].

Taken together, these findings suggest the best product strategy is not to chase a single monolithic “AI super-app” immediately, but to launch a focused MVP with:

official-source ingestion,

citation-grounded campus Q&A,

summarization,

independent spam filtering,

human oversight and privacy controls [1] [33] [42] [45].

Overall confidence: High - The architectural direction is well supported across multiple source clusters, though operational details like unified authentication, analytics, and multi-tenant deployment remain underdocumented in the provided evidence [38] [39] [40].

References

Kumar, K.V.S.R.D.S.P., Thota, S., Reddy, Y.C.A.P., (...), Pinagani, S. (2026) AI-Powered College Chatbot with Agentic Multi-Model RAG. *2026 International Conference on Computing Theory and Wireless Communications, ICCTWC 2026*. https://doi.org/10.1109/ICCTWC68241.2026.11583133

Allen, M., Naeem, U., Gill, S.S. (2024) Q-Module-Bot: A Generative AI-Based Question and Answer Bot for Module Teaching Support. *IEEE Transactions on Education*. https://doi.org/10.1109/TE.2024.3435427

Jamil, A., Azhar, M., Amjad, A., (...), Usman, M. (2025) Text Summarization and Automatic Clustering of Scanned Documents by Using Hybrid Deep Learning Algorithms. *3rd International Conference on Emerging Research in Computational Science, ICERCS 2025*. https://doi.org/10.1109/ICERCS65898.2025.11580800

Boczar, A., Boczar, J., Mi, X., (...), Tolbert, J. (2023) A case study on migrating large digital collections into a consolidated repository. *Journal of Academic Librarianship*. https://doi.org/10.1016/j.acalib.2023.102685

Mohan Raj, T., Visvasa Raja, S. (2025) CampusIssueMiner: Privacy-Preserving Triage of Code-Mixed Campus Grievances. *Proceedings of 5th International Conference on Evolutionary Computing and Mobile Sustainable Networks, ICECMSN 2025*. https://doi.org/10.1109/ICECMSN68058.2025.11383400

Dayata, W.M.A., Yap, S.Y.C. (2024) Carolinian Chatbot: End-to-End Question-Answer Pipeline on the University of San Carlos Policies, Procedures, and Guidelines Via Retrieval Augmented Generation. *COMNETSAT 2024 - IEEE International Conference on Communication, Networks and Satellite*. https://doi.org/10.1109/COMNETSAT63286.2024.10862750

Mushtaq, F., Azam, F., Anwar, M.W. (2024) Performance Comparison of Single Code Base Development Tools: Flutter, React Native, and Xamarin. *Proceedings - 2024 14th International Conference on Software Technology and Engineering, ICSTE 2024*. https://doi.org/10.1109/ICSTE63875.2024.00011

Majchrzak, T.A., Biørn-Hansen, A., Grønli, T.-M. (2018) Progressive web apps: The definite approach to cross-platform development?. *Proceedings of the Annual Hawaii International Conference on System Sciences*. https://www.scopus.com/pages/publications/85085511705

Thao, C.V.T., Tam, T.D., Le, C., (...), Tran, T. (2025) Security Techniques in Cross-Platform Software Development Based on the Flutter Framework. *EEE-AM 2025 - IEEE 2nd Asia Meeting on Environment and Electrical Engineering*. https://doi.org/10.1109/EEE-AM66675.2025.11473565

Rajaguru, S., Chathumini, L., Kumara, S., Wijerathna, A. (2025) Enhancing Flutter App Development: Addressing Configuration and Compatibility Bugs Using Large Language Models. *Proceedings - International Research Conference on Smart Computing and Systems Engineering, SCSE 2025*. https://doi.org/10.1109/SCSE65633.2025.11030993

Pondel, M., Chomiak-Orsa, I., Sobińska, M., (...), Grenczuk, A. (2026) Retrieval-Augmented Generation for Organisational Communication: Designing and Deploying an AI Assistant for University-Student Interaction. *IFIP Advances in Information and Communication Technology*. https://doi.org/10.1007/978-3-032-18920-2_6

Ahmed, S., Shaheen, R., Khaled, A.S., (...), Shakeel, M.U. (2026) Enhancing Academic Support with AI: A Case Study of Chatbot Integration Using Microsoft Copilot Studio at Jazan University. *International Conference on Artificial Intelligence, Computer, Data Sciences, and Applications, ACDSA 2026*. https://doi.org/10.1109/ACDSA67686.2026.11468064

Lewis, P., Perez, E., Piktus, A., (...), Kiela, D. (2020) Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*. https://www.scopus.com/pages/publications/85108449607

Radaeva, A., Stepanov, N., Suleykin, A., (...), Egorova, K. (2026) A Hybrid Multi-level Retrieval Architecture for an AI Assistant in University Admissions Support. *Proceedings - 2026 6th International Conference on Technology Enhanced Learning in Higher Education, TELE 2026*. https://doi.org/10.1109/TELE71748.2026.11591353

Srinath, G.M., Gowda, K.N., Mahesh, V., (...), Prajwal, S. (2026) Campus AI RAG: A Multi-LLM Retrieval-Augmented Generation System for College Information Support. *ICKECS 2026 - Proceedings of 4th IEEE International Conference on Knowledge Engineering and Communication Systems*. https://doi.org/10.1109/ICKECS70176.2026.11527960

Yang, L., Li, Y. (2026) PTMR-ECM: Physical topology-constrained and temporal evolution-aware multimodal retrieval and question answering framework for engineering change management. *Advanced Engineering Informatics*. https://doi.org/10.1016/j.aei.2026.105154

Yang, S., Wu, A., Wu, X., (...), He, L. (2026) Evidence-chain-driven multimodal retrieval question answering. *Knowledge-Based Systems*. https://doi.org/10.1016/j.knosys.2026.116358

Comiti, G., Bisgambiglia, P.-A., Lameta, N. (2026) A hybrid quantitative–qualitative RAG approach for territorial well-being decision support. *Machine Learning with Applications*. https://doi.org/10.1016/j.mlwa.2026.101005

Roy, T., Kumar, A., Raghuvanshi, D., (...), Tondulkar, R. (2024) SciSpace Copilot: Empowering Researchers through Intelligent Reading Assistance. *Proceedings of the AAAI Conference on Artificial Intelligence*. https://doi.org/10.1609/aaai.v38i21.30578

Lingling, W., Shijie, S., Xueyi, L. (2026) From automation to augmentation: a retrieval-augmented large language model framework for reshaping learning experiences in higher education. *Frontiers in Psychology*. https://doi.org/10.3389/fpsyg.2026.1870870

Muthukumar, D.S., Vendhan, D., Achari, P.H., (...), Kumar, P.Y. (2026) Smart College Helpdesk Assistant Using Machine Learning & NLP. *Proceedings of 8th International Conference on Intelligent Sustainable Systems, ICISS 2026*. https://doi.org/10.1109/ICISS67859.2026.11453598

Hemaswathi, S., TamilMuthu, K., Hariharasuthan, K.B., (...), Sarveen Bala, R. (2026) Next - Gen University Helpdesk (NexU). *2026 IEEE International Conference on Emerging Computing and Intelligent Technologies, ICoECIT 2026*. https://doi.org/10.1109/ICoECIT68303.2026.11497274

Zhang, J., Liu, Q. (2026) Design of an Intelligent Question Answering System for Universities Integrating Knowledge Graphs and Large Language Models. *Proceedings of the 3rd International Conference on Educational Digitalization and Computer Science in the Guangdong-Hong Kong-Macao Greater Bay Area, EDCS 2026*. https://doi.org/10.1145/3828183.3828497

Iddrisu, W.A., Adjei-Gyabaa, S.K., Akoto, I. (2023) Content-Based Spam Classification of Academic E-mails: A Machine Learning Approach. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-981-19-9888-1_7

Alqatawna, J., Bilquise, G., Al-Zoubi, A.M. (2024) Optimizing Email Spam Detection in Academic Environments: A Machine Learning Approach. *Proceedings - 2024 International Conference on Artificial Intelligence, Metaverse and Cybersecurity, ICAMAC 2024*. https://doi.org/10.1109/ICAMAC62387.2024.10829236

Kalluri, S.R., Billa, D.R., Chinthala, A.R., Rana, M.M. (2025) EmailSpam ML : Optimized ML Algorithms for Spam Email Detection, Performance Insights, and Future Research Directions. *International Conference on Advanced Communication Technology, ICACT*. https://doi.org/10.23919/ICACT63878.2025.10936680

Bacanin, N., Zivkovic, M., Stoean, C., (...), Strumberger, I. (2022) Application of Natural Language Processing and Machine Learning Boosted with Swarm Intelligence for Spam Email Filtering. *Mathematics*. https://doi.org/10.3390/math10224173

Mohd, R., Ong, Z.Z. (2025) A Study on LLMs for Spam Email Classification. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-3-031-84460-7_45

Singh, B., Mishra, R., Nagina, K., (...), Khan, N.R. (2026) Real-Time Unified Spam Detection System for Email and SMS. *2026 2nd International Conference on Big Data and Machine Learning, ICBDML 2026*. https://doi.org/10.1109/ICBDML68582.2026.11544748

Ejirika, E.R., Omotehinwa, T.O. (2024) Analysis of Machine Learning Models for Spam Email Detection and Real-Time Integration. *International Conference on Science, Engineering and Business for Driving Sustainable Development Goals, SEB4SDG 2024*. https://doi.org/10.1109/SEB4SDG60871.2024.10630209

Saeed, A.Q., Aldulaimi, M.H., Ismail, I.A., (...), Ghazal, T.M. (2024) Integrating Three Machine Learning Algorithms in Ensemble Learning Model for Improving Content-based Spam Email Recognition. *Journal of Soft Computing and Data Mining*. https://doi.org/10.30880/jscdm.2024.05.02.014

Canay, Ö., Kocabıçak, Ü. (2023) An innovative data collection method to eliminate the preprocessing phase in web usage mining. *Engineering Science and Technology, an International Journal*. https://doi.org/10.1016/j.jestch.2023.101360

Li, C., Chen, G., Li, F., (...), Wu, Y. (2025) Decoding Consumer-Perceived Risks in China's C-end Online Purchasing Pre-made Dishes: A Quality Safety Risk Identification Model Based on Grounded Theory. *Journal of Food Protection*. https://doi.org/10.1016/j.jfp.2025.100522

Krause, T., Uetz, R., Kretschmann, T. (2019) Recognizing Email Spam from Meta Data only. *2019 IEEE Conference on Communications and Network Security, CNS 2019*. https://doi.org/10.1109/CNS.2019.8802827

Hu, Y., Guo, C., Ngai, E.W.T., (...), Chen, S. (2010) A scalable intelligent non-content-based spam-filtering framework. *Expert Systems with Applications*. https://doi.org/10.1016/j.eswa.2010.05.020

Cohen, A., Nissim, N., Elovici, Y. (2018) Novel set of general descriptive features for enhanced detection of malicious emails using machine learning methods. *Expert Systems with Applications*. https://doi.org/10.1016/j.eswa.2018.05.031

Nikiforova-Ilieva, K., Georgiev, T. (2026) Architectural Frameworks and Strategic Selection of AI-Driven Administrative Chatbots in Educational Environments. *2026 25th International Symposium INFOTEH-JAHORINA, INFOTEH 2026 - Proceedings*. https://doi.org/10.1109/INFOTEH68759.2026.11477638

Coughlan, T., Iniesto, F. (2025) What should I know? Analysing behaviour and feedback from student use of a virtual assistant to share information about disabilities. *Internet and Higher Education*. https://doi.org/10.1016/j.iheduc.2025.101002

Garrison, E., MacNeil, S., Hantula, D.A., (...), Vučetić, S. (2025) Exploring the challenges and assistive technology for autistic job seekers across employment pathways. *Research in Developmental Disabilities*. https://doi.org/10.1016/j.ridd.2025.105155

Atusingwize, Edwinah, Nilsson, Maria, Egan Sjölander, Annika, (...), Landstedt, Evelina (2024) Exploring links—exposure to alcohol adverts on social media in relation to alcohol use among university students in Uganda. *Alcohol and Alcoholism*. https://doi.org/10.1093/alcalc/agae081

Nie, Yu, Fu, Jianming, Lv, Xinghang, (...), Peng, Guojun (2025) WAScope: Detecting privacy data leakage with web application-specific API confusion. *Alexandria Engineering Journal*. https://doi.org/10.1016/j.aej.2025.08.006

Kasneci, E., Sessler, K., Küchemann, S., (...), Kasneci, G. (2023) ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and Individual Differences*. https://doi.org/10.1016/j.lindif.2023.102274

Cotton, D.R.E., COTTON, P.A., Shipway, J.R. (2024) Chatting and cheating: Ensuring academic integrity in the era of ChatGPT. *Innovations in Education and Teaching International*. https://doi.org/10.1080/14703297.2023.2190148

Michel-Villarreal, R., Vilalta-Perdomo, E., Salinas-Navarro, D.E., (...), Gerardou, F.S. (2023) Challenges and Opportunities of Generative AI for Higher Education as Explained by ChatGPT. *Education Sciences*. https://doi.org/10.3390/educsci13090856

Manolache, F.B., Rusu, O., Adusumilli, P. (2008) Spam handling in heterogeneous environments. *7th RoEduNet International Conference, RoEduNet 2008*. https://www.scopus.com/pages/publications/84907061190

Omotehinwa, T.O., Oyewola, D.O. (2023) Hyperparameter Optimization of Ensemble Models for Spam Email Detection. *Applied Sciences (Switzerland)*. https://doi.org/10.3390/app13031971

Atwe, L., Jazzar, M., Eleyan, A., Bejaoui, T. (2025) A Comparative Study: Using Machine Learning and Transformers Model to Identify Spam in SMS. *2025 7th International Conference on Smart Applications, Communications and Networking, SmartNets 2025*. https://doi.org/10.1109/SmartNets65254.2025.11106848

https://www.sciencedirect.com/science/article/pii/S2661890726001535

Jadeja, M.V., Lad, M.J., Kamboj, P. (2025) Optimizing Response Precision: A Dual-Model Chatbot Using RAG & NLP. *2025 World Skills Conference on Universal Data Analytics and Sciences, WorldSUAS 2025*. https://doi.org/10.1109/WorldSUAS66815.2025.11199058

Deng, X., Yang, D., Zhang, Y. (2024) Intelligent Orientation Robot Based on Large Language Models and Retrieval-Augmented Generation. *2024 5th International Conference on Artificial Intelligence and Computer Engineering, ICAICE 2024*. https://doi.org/10.1109/ICAICE63571.2024.10863858

Mayor-Alonso, E., Vidal, J., Rodríguez-Esteban, A. (2026) Evaluating the response quality of LLM-driven university guidance applications. *Revista de Educacion*. https://doi.org/10.4438/1988-592X-RE-2025-411-729

Caballero Castellanos, J.R., Cardona, M., Ordoñez-Ávila, J.L. (2023) Chatbot for Validation of Research Topics for Engineering Students. *Proceeding of the 2023 IEEE 41st Central America and Panama Convention, CONCAPAN XLI 2023*. https://doi.org/10.1109/CONCAPANXLI59599.2023.10517536

Sobhana, M., Yamini, A., Hindu, K., Narayana, Y.L. (2023) Navbot—College Navigation Chatbot Using Deep Neural Network. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-981-19-5845-8_37

Reddy, B., Geethika, E., Akshaya, E. (2026) Indoor Navigation: Personalized and Context Aware Campus Directions. *Cognitive Science and Technology*. https://doi.org/10.1007/978-981-95-0144-1_35

Wilkinson, M.D., Dumontier, M., Aalbersberg, I.J., (...), Mons, B. (2016) Comment: The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*. https://doi.org/10.1038/sdata.2016.18

Hai, R., Koutras, C., Quix, C., Jarke, M. (2023) Data Lakes: A Survey of Functions and Systems. *IEEE Transactions on Knowledge and Data Engineering*. https://doi.org/10.1109/TKDE.2023.3270101

Bonino da Silva Santos, L.O.B., Burger, K., Kaliyaperumal, R., Wilkinson, M.D. (2023) FAIR Data Point: A FAIR-Oriented Approach for Metadata Publication. *Data Intelligence*. https://doi.org/10.1162/dint_a_00160

Albertoni, R., Browning, D., Cox, S., (...), Winstanley, P. (2024) The W3C Data Catalog Vocabulary, Version 2: Rationale, Design Principles, and Uptake. *Data Intelligence*. https://doi.org/10.1162/dint_a_00241

Li, Z., Prabhu, S.P., Popp, Z.T., (...), Chen, J. (2025) A natural language processing approach to support biomedical data harmonization: Leveraging large language models. *PLOS ONE*. https://doi.org/10.1371/journal.pone.0328262

Emanuel, M. (2021) Adventures in Migrating Massive Archival Collections to Digital Commons. *Open Information Science*. https://doi.org/10.1515/opis-2021-0007

König, René, Uphues, Steffen, Vogt, Verena, Kolany-Raiser, Barbara (2020) The practical and ethical challenges in acquiring and sharing digital trace data: Negotiating public-private partnerships. *New Media & Society*. https://doi.org/10.1177/1461444820924622

Sadaraboina, P.T., Puram, S. (2026) On-Device Generative AI Integration In Android Applications: Architectural Patterns For Scalable And Efficient AI Systems. *International Journal of Artificial Intelligence and Machine Learning*. https://doi.org/10.51483/IJAIML.6.6s.2026.1031-1040

Đambić, G., Maurovic, A., Ogrizek Biškupić, I., Radovan, A. (2026) Comparative Evaluation of AI Programming Assistants: An Exploratory Longitudinal Case Study of GitHub Copilot, ChatGPT, and Cursor Configurations in Full-Stack Development. *Information (Switzerland)*. https://doi.org/10.3390/info17090918

Bento, A.C., Pereira-Barretto, M.R., Reinaldo Silva, J., (...), Torres-Torres, E.Y. (2025) Integrating AI Development Tools in the Classroom: A Multi-Tool Case Study on Learning Outcomes and Code Quality. *2025 2nd International Conference on Artificial Intelligence and Teacher Education, ICAITE 2025*. https://doi.org/10.1109/ICAITE68636.2025.11442616

Raveendran, K., Karuneswaran, G., Parthipan, V., (...), Ragel, R.G. (2025) RAG SLIITbot: Sri Lankan Universities Develop AI Tutor Tailored to Unique Teaching Styles. *ICIIS 2025 - Next-Gen Engineering for Industry 5.0: Innovating Intelligent Systems for Human Centric Future: Proceedings of 2025 IEEE 19th International Conference on Industrial and Information Systems*. https://doi.org/10.1109/ICIIS69028.2026.11450883

Ramli, A.A.R.B., Krathu, W., Bramantoro, A. (2025) PUENYAI Bot: A Retrieval-Augmented LLM Chatbot for University Information Services. *2025 7th International Conference on Applied Computational Intelligence in Information Systems: Intelligent and Resilient Digital Innovations for Sustainable Living, ACIIS 2025 - Proceedings*. https://doi.org/10.1109/ACIIS66255.2025.11402978

Li, Z., Wang, Z., Wang, W., (...), Wang, F.L. (2025) Retrieval-augmented generation for educational application: A systematic survey. *Computers and Education: Artificial Intelligence*. https://doi.org/10.1016/j.caeai.2025.100417

Guettala, M., Bourekkache, S., Okba, O., Harous, S. (2026) Generative AI and cloud deployment for intelligent systems: a retrieval-augmented generation framework using amazon bedrock. *Information Processing and Management*. https://doi.org/10.1016/j.ipm.2026.104837

Joseph, J., Thomas, B., Pathak, N., Sharma, N. (2026) SpamXFormer: A Hybrid Transformer-CNN Model with Attention for Enhanced Email Spam Detection. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-3-032-25958-5_31

Kadam, K.V., Rohokale, V.M. (2022) A hybrid meta-heuristic-based multi-objective feature selection with adaptive capsule network for automated email spam detection. *International Journal of Intelligent Robotics and Applications*. https://doi.org/10.1007/s41315-021-00217-9

He, Z., Liu, C., Hu, B., Jia, D. (2026) SpamLLM: Leveraging Large Language Models for Robust Spam Email Classification. *International Journal of Intelligent Systems*. https://doi.org/10.1155/int/2309553

Breiman, L. (2001) Random forests. *Machine Learning*. https://doi.org/10.1023/A:1010933404324

Androutsopoulos, Ion, Koutsias, John, Chandrinos, Konstantinos V., Spyropoulos, Constantine D. (2000) Experimental comparison of Naive Bayesian and keyword-based anti-spam filtering with personal e-mail messages. *SIGIR Forum (ACM Special Interest Group on Information Retrieval)*. https://doi.org/10.1145/345508.345569

Barushka, A., Hajek, P. (2018) Spam filtering using integrated distribution-based balancing approach and regularized deep neural networks. *Applied Intelligence*. https://doi.org/10.1007/s10489-018-1161-y

Chan, C.K.Y., Hu, W. (2023) Students' voices on generative AI: perceptions, benefits, and challenges in higher education. *International Journal of Educational Technology in Higher Education*. https://doi.org/10.1186/s41239-023-00411-8

Jeon, J., Lee, S. (2023) Large language models in education: A focus on the complementary relationship between human teachers and ChatGPT. *Education and Information Technologies*. https://doi.org/10.1007/s10639-023-11834-1